Boolean Algebra and Logic Gates- Part 2

K. Sridharan

Review and beyond

- We have so far discussed about some aspects of Boolean logic
- We indicated that the notion of truth table can also be used to prove theorems/relations in binary logic.
- We will discuss about truth tables now.

What is a truth table ?

- A truth table is a way of defining logical operations. For example, the truth tables of 2-input AND, 2-input OR and NOT are:

AND

x	y	$x \cdot y$
0	0	0
0	1	0
1	0	0
1	1	1

OR

x	y	$x + y$
0	0	0
0	1	1
1	0	1
1	1	1

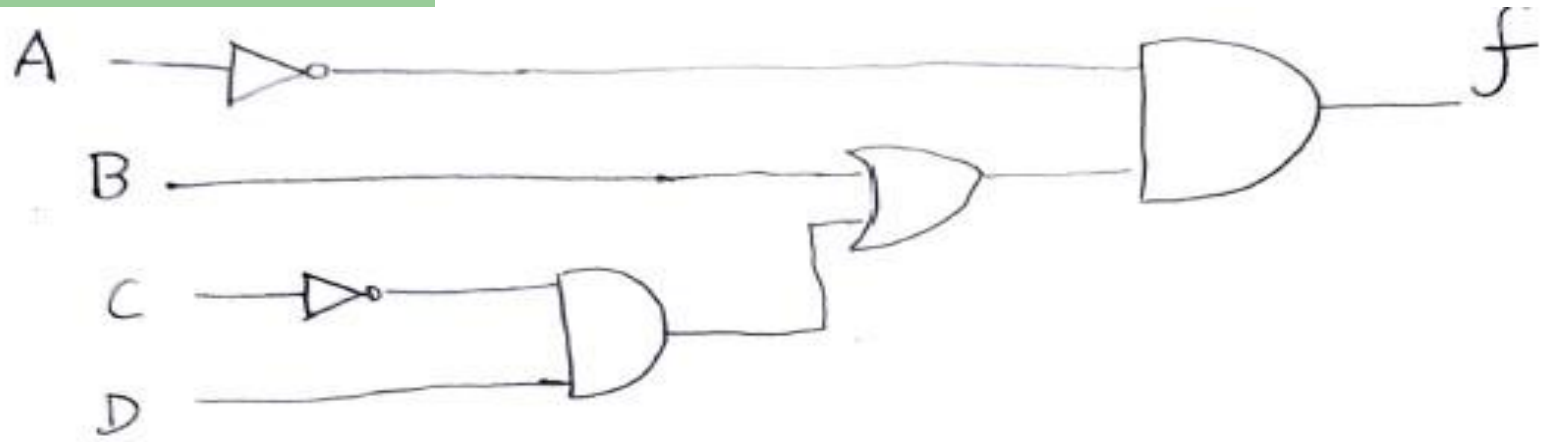
NOT

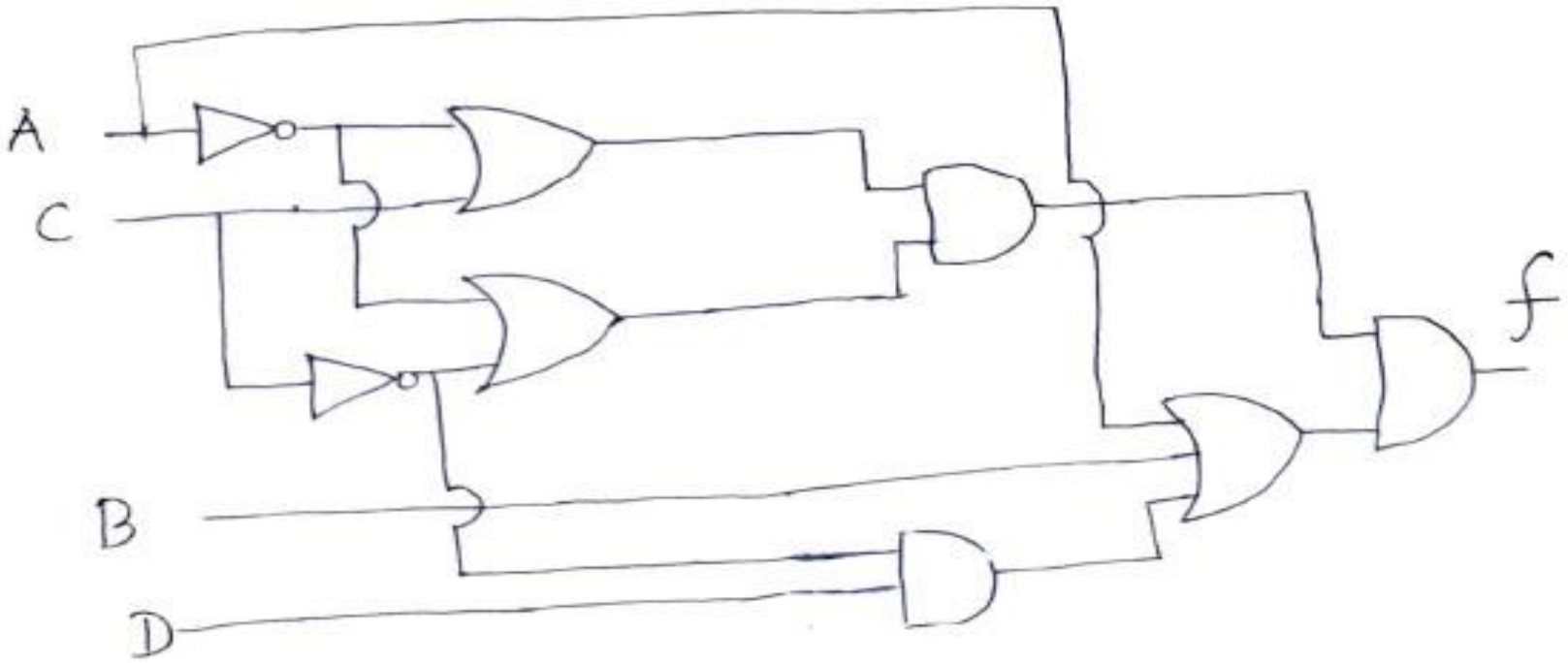
x	x'
0	1
1	0

How are truth tables useful ?

- What we have given so far are truth tables of Boolean functions AND, OR and NOT.
- The tables can also be used to prove the equivalence of Boolean expressions.
- We will consider the circuit presented earlier once more.

Consider these circuits seen earlier



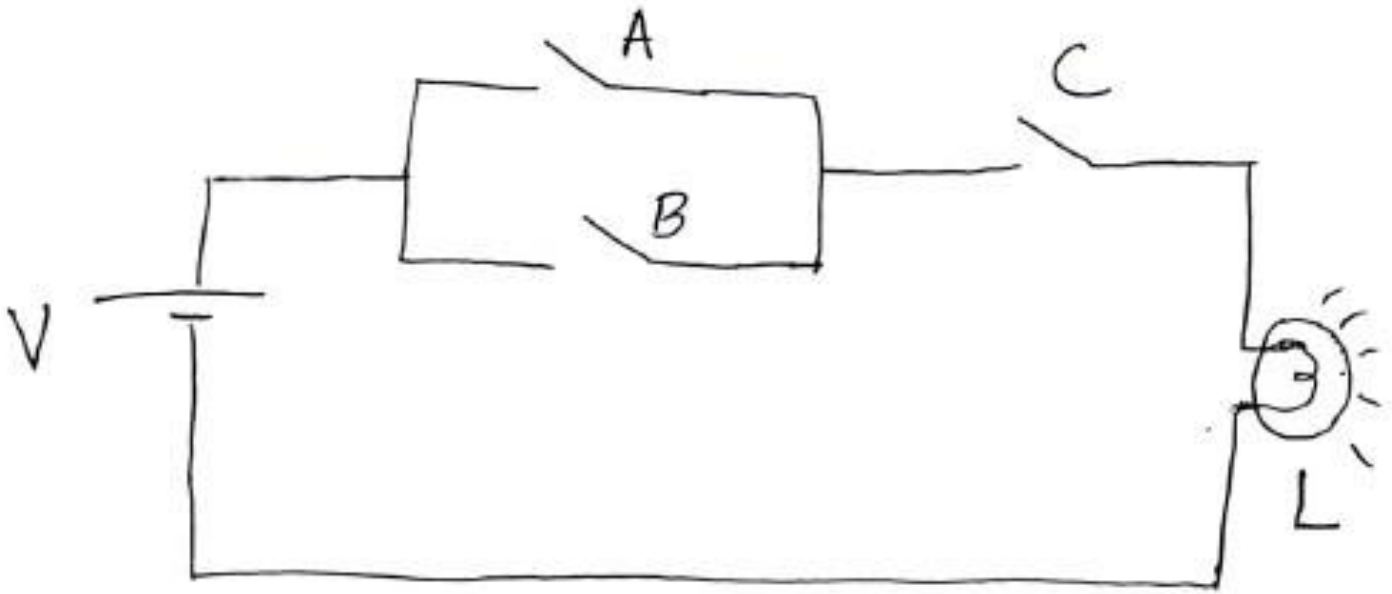


Exercise for students

- Write the truth tables for the function f in both the circuits and observe the contents of the table.

Is there any other use of a truth table ?

Consider the circuit given below:



Truth table describing the circuit behaviour

A	B	C	L
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

$$L = \bar{A}BC + A\bar{B}C + ABC$$

$$= C(A+B)$$

How did we simplify to C(A+ B) ?

- Note that L = A’BC + AB’C + ABC can be rewritten as:

$$L = A'BC + AB'C + ABC + ABC$$

(introducing ABC)

$$= A'BC + ABC + AB'C + ABC$$

$$= (A' + A)\,BC + AC\,(B + B')$$

$$= BC + AC$$

$$= C\,(A + B)$$

Minterms

- Two binary variables, x and y, can be combined via an AND operation. It can lead to four possible combinations, namely x'y', x'y, xy' and xy. Each of these is called a minterm (or standard product).

How do we remember the word “Minterm” ?

- One way to remember this is by noting that AND itself is a minimum operation. That is, AND(x,y) = min(x,y) where x and y are binary variables that take values of 0 or 1.

How many minterms exist for n variables ?

- Note that each variable appears either as itself or as its complement in a minterm.
- We will have 2^n minterms for n variables

Maxterms

- Two binary variables, x and y, can also be combined via an OR operation. It can lead to four possible combinations, namely x’ + y’, x’ + y, x + y’ and x+y. Each of these is called a maxterm (or standard sum).
- One way to remember maxterm is by noting that OR itself is a max operation. That is, OR(x,y) = max(x,y) where x and y are binary variables that take values of 0 or 1.
- We have 2^n maxterms for n variables.

Boolean function as the sum of minterms

- Consider the function for lighting a bulb in the circuit shown earlier: L = A’BC + AB’C + ABC
- This is called as sum of minterms. Note sum here refers to OR.
- We may also refer to this as sum of products (SOP).
- An alternate way to express L is as the product (AND) of maxterms. We discuss this next.

Same Boolean function as the product of maxterms

- First consider the complement of the Boolean function, namely L'. This is obtained by forming a minterm for each combination that produces a 0 and ORing the terms.
- To extract the terms leading to a 0, can use the truth table.
- So we have L' = A'B'C' + A'B'C + A'BC' + AB'C' + ABC'.

More on expressing the function as the product of maxterms

- Having got L', we need to take its complement again !
- That is, take the complement of L' to get L, namely

  (A+B+C)(A+B+C')(A+B'+C)(A'+B+C)(A'+B'+C)

- We have omitted the dots in between

Canonical form

- Boolean functions expressed as the sum of minterms or as the product of maxterms are said to be in canonical form.
- Note the use of DeMorgan's theorem while taking the complement of L' to obtain L.
- We may represent sum by $\sum$ and product by $\prod$

Verify (Exercise for students)

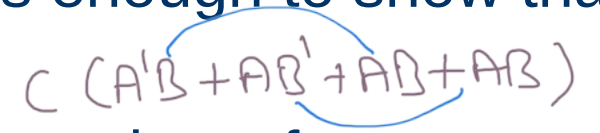
- Show that the product of maxterms given by

(A+B+C)(A+B+C')(A+B'+C)(A'+B+C)(A'+B'+C)

is indeed the same as the sum of minterms,

i.e., A'BC + AB'C + ABC

- Note that it is enough to show that both reduce to C(A + B)



- Sometimes, product of maxterms may be called product of sums (POS).

How to derive all minterms for a Boolean function ?

- Given a Boolean function such as C(A + B), just write the truth table for the function and extract the minterms (those terms for which the function value is 1). Note that each minterm should have all the variables (either as such or as the complement).
- Sometimes, the sum of minterms may also be written in a form like this: $\sum$ (3,5,7)

Textbook and Reference

- M. Mano and M. Ciletti, Digital Design, Pearson Education, 2018
- J. Wakerly, Digital Design – Principles & Practices, Pearson Education, 2008